

1. _____ are made with the intent of accomplishing tasks which either too tedious for humans or are time-consuming.

- a. Machines
- b. Gadgets
- c. Equipment
- d. Tools

Ans- Machines

2. _____ refers to man-made does not occur naturally.

- e. Intelligence
- f. Artificial
- g. Natural
- h. Native

Ans- Artificial

3. _____ refers to the ability to perceive or infer information, and to retain it as knowledge to be applied toward adaptive behaviors within an environment or context.

- i. Behavior
- j. Nature
- k. Habits
- l. Intelligence

Ans- Intelligence

4..Ms. Rani is intelligent in performing calculations and logical reasoning. This ability refers to

- a. Linguistic Intelligence
- b. Spatial Visual Intelligence
- c. Mathematical Intelligence
- d. Existential Intelligence

Ans- Mathematical Intelligence

5. Mr. Anurag is excellent at publishing articles on his blog. This ability refers to

- a. Linguistic Intelligence
- b. Spatial Visual Intelligence
- c. Mathematical Intelligence
- d. Existential Intelligence

Ans- Linguistic Intelligence

6. The ability of religious and spiritual awareness is related to which of the following intelligences?

- a. naturalist intelligence
- b. interpersonal intelligence
- c. existential intelligence
- d. kinaesthetic intelligence

Ans- existential intelligence

7. The ability to communicate with others by understanding other people's feelings & influence on the person is known as

- a. naturalist intelligence
- b. interpersonal intelligence
- c. existential intelligence
- d. kinaesthetic intelligence

Ans- interpersonal intelligence

8. Speech recognition, Image recognition, and the ability to take action are examples of

- a. to perceive, understand, and act
- b. reasoning and planning

- c. learning and adaption
- d. none of these

Ans- to perceive, understand, and act

9. What is the field of Natural Language Processing (NLP)?

- a) Computer Science
- b) Artificial Intelligence
- c) Linguistics
- d) All of the mentioned

Ans- All of the mentioned

10. Which of the following includes major tasks of NLP?

- a) Automatic Summarization
- b) Discourse Analysis
- c) Machine Translation
- d) All of the mentioned

Ans- All of the mentioned

11. What is the main challenge/s of NLP?

- a) Handling Ambiguity of Sentences
- b) Handling Tokenization
- c) Handling POS-Tagging
- d) All of the mentioned

Ans- Handling Ambiguity of Sentences

12. Which algorithm is often used for text classification tasks in NLP?

- a. Naive Bayes
- b. Decision Trees
- c. K-means clustering
- d. Principal Component Analysis (PCA)

Ans- Naive Bayes

13. What is “word sense disambiguation” in NLP?

- a. The process of determining the correct meaning of a word based on context
- b. The process of identifying named entities in text
- c. The process of tokenizing text into individual words
- d. The process of translating text into another language

Ans- The process of determining the correct meaning of a word based on context

14. What is Artificial Intelligence?

- a) Artificial Intelligence is a field that aims to make humans more intelligent
- b) Artificial Intelligence is a field that aims to improve the security
- c) Artificial Intelligence is a field that aims to develop intelligent machines
- d) Artificial Intelligence is a field that aims to mine the data

Answer: Artificial Intelligence is a field that aims to develop intelligent machines

15. Who is the inventor of Artificial Intelligence?

- a) Geoffrey Hinton
- b) Andrew Ng
- c) John McCarthy
- d) Jürgen Schmidhuber

Answer: John McCarthy

16. Which of the following is the branch of Artificial Intelligence?

- a) Machine Learning
- b) Cyber forensics
- c) Full-Stack Developer
- d) Network Design

Answer: Machine Learning

17. What is the goal of Artificial Intelligence?

- a) To solve artificial problems
- b) To extract scientific causes
- c) To explain various sorts of intelligence
- d) To solve real-world problems

Answer: To explain various sorts of intelligence

18. Which of the following is an application of Artificial Intelligence?

- a) It helps to exploit vulnerabilities to secure the firm
- b) Language understanding and problem-solving (Text analytics and NLP)
- c) Easy to create a website
- d) It helps to deploy applications on the cloud

Answer: Language understanding and problem-solving (Text analytics and NLP)

19. In how many categories process of Artificial Intelligence is categorized?

- a) categorized into 5 categories
- b) processes are categorized based on the input provided
- c) categorized into 3 categories
- d) process is not categorized

Answer: categorized into 3 categories

20. Based on which of the following parameter Artificial Intelligence is categorized?

- a) Based on functionally only
- b) Based on capabilities only
- c) Based on capabilities and functionally
- d) It is not categorized

Answer: Based on capabilities and functionally

21. Which of the following is a component of Artificial Intelligence?

- a) Learning
- b) Training
- c) Designing
- d) Puzzling

Answer: Learning

22. What is the function of an Artificial Intelligence "Agent"?

- a) Mapping of goal sequence to an action
- b) Work without the direct interference of the people
- c) Mapping of precept sequence to an action
- d) Mapping of environment sequence to an action

Answer: Mapping of precept sequence to an action

23. Which of the following is not a type of Artificial Intelligence agent?

- a) Learning AI agent

- b) Goal-based AI agent
- c) Simple reflex AI agent
- d) Unity-based AI agent

Answer: Unity-based AI agent

24.. What is Natural Language Processing (NLP) primarily concerned with?

- a) Analyzing and understanding human languages
- b) Improving network communication speeds
- c) Developing new programming languages
- d) Enhancing graphical user interfaces

Answer: Analyzing and understanding human languages

25. Which of the following is a common application of NLP?

- a) Image recognition
- b) Sentiment analysis
- c) Game development
- d) Cryptography

Answer: Sentiment analysis

26. Tokenization in NLP refers to:

- a) Encrypting sensitive text data
- b) Breaking down text into individual words or phrases
- c) Translating text from one language to another
- d) Storing large volumes of text data

Answer: Breaking down text into individual words or phrases

27. What is a "corpus" in the context of NLP?

- a) A set of rules for grammar and syntax

- b) A large and structured set of texts
- c) A programming language for NLP
- d) An algorithm for text classification

Click to View Answer and Explanation

Answer: A large and structured set of texts

28. Which algorithm is commonly used for word embeddings in NLP?

- a) Decision Trees
- b) Naive Bayes Classifier
- c) Word2Vec
- d) K-Means Clustering

Answer: Word2Vec

29. Named Entity Recognition (NER) in NLP is used for:

- a) Correcting grammar in sentences
- b) Identifying and categorizing key information in text
- c) Translating text between languages
- d) Compressing text data for efficient storage

Answer: Identifying and categorizing key information in text

30. What does "stemming" in NLP involve?

- a) Visualizing text data
- b) Removing affixes from words to get the root form
- c) Encrypting text data for secure transmission
- d) Generating synthetic speech

Answer: Removing affixes from words to get the root form

31. The Turing Test is used to assess:

- a) A computer's ability to exhibit intelligent behavior
- b) The speed of text processing algorithms
- c) The accuracy of speech recognition systems
- d) The efficiency of database storage for linguistic data

Answer: A computer's ability to exhibit intelligent behavior

32. What is the purpose of "Part-of-Speech tagging" in NLP?

- a) To identify the part of speech of each word in a sentence
- b) To correct misspellings in a text
- c) To encrypt text messages
- d) To count the number of words in a text

Answer: To identify the part of speech of each word in a sentence

33. A "chatbot" in NLP is designed to:

- a) Store and retrieve large databases of text
- b) Simulate conversational interaction with humans
- c) Translate text between different programming languages
- d) Perform image recognition tasks

Answer: Simulate conversational interaction with humans

34. What does NLTK stand for?

- a) Natural Language Text Kit
- b) Natural Language Toolkit
- c) Neural Learning Toolkit
- d) Natural Language Technical Kit

Answer: Natural Language Toolkit

35. Which of the following is NOT a module in NLTK?

- a) nltk.tokenize
- b) nltk.corpus
- c) nltk.parser
- d) nltk.machine_learning

Answer: nltk.machine_learning

36. What is the primary function of the nltk.tokenize module?

- a) To parse sentences into syntactic trees
- b) To convert text into numerical data
- c) To break text into words or sentences
- d) To remove stop words from text

Answer: To break text into words or sentences

37. Which of the following corpus is included in NLTK?

- a) Shakespeare's Complete Works
- b) Brown Corpus
- c) Wikipedia Corpus
- d) Google News Corpus

Answer: Brown Corpus

38. How do you download a corpus in NLTK?

- a) nltk.download('corpus_name')
- b) nltk.corpus.download('corpus_name')
- c) nltk.load_corpus('corpus_name')
- d) nltk.get_corpus('corpus_name')

Answer: nltk.download('corpus_name')

39. Which of the following methods in NLTK is used to find the frequency distribution of words?

- a) nltk.FreqDist()
- b) nltk.WordFreq()
- c) nltk.frequency()
- d) nltk.Text()

Answer: nltk.FreqDist()

40. What does the nltk.pos_tag() function do?

- a) Converts text to lowercase
- b) Splits a sentence into words
- c) Tags words in a sentence with their part of speech
- d) Extracts named entities from a sentence

Answer: Tags words in a sentence with their part of speech

41. Which of the following is a stopwords list available in NLTK?

- a) nltk.stopwords.words('english')
- b) nltk.remove_stopwords('english')
- c) nltk.get_stopwords('english')
- d) nltk.stopwords('english')

Answer: nltk.stopwords.words('english')

42. What does the nltk.ne_chunk() function do?

- a) Identifies named entities in a text
- b) Breaks text into chunks of words
- c) Generates a word cloud
- d) Segments a document into sentences

Answer: Identifies named entities in a text

43. Which of the following methods can be used for stemming in NLTK?

- a) nltk.stem.PorterStemmer()
- b) nltk.stem.LancasterStemmer()
- c) nltk.stem.SnowballStemmer()
- d) All of the above

Answer: All of the above

44. A _____ is a sequence of observations over a certain period.

- A. Theano
- B. Scikit
- C. Time series
- D. Tensor

Ans : Time series

45. A time series has _____ components.

- A. 1
- B. 2
- C. 3
- D. 4

Ans : 4

46. Which of the following is the cyclic behavior of time series?

- A. Level
- B. Trend
- C. Seasonality
- D. Noise

Ans : Seasonality

47. LSTM stands for?

- A. Long Short-Term Model
- B. Long Short-Term Memory model
- C. Long Short Memory model
- D. LS Memory model

Ans : Long Short-Term Memory model

48. Level is the mean value around which the series varies.

- A. TRUE
- B. FALSE
- C. Can be true or false
- D. Can not say

Ans : TRUE

49. _____ is a technique for smoothing univariate time-series by assigning exponentially decreasing weights to data over a time period.

- A. Walk Forward Validation
- B. Prophet Model
- C. LSTM Model
- D. Exponential Smoothing

Ans : Exponential Smoothing

50. Facebook open sourced the prophet model which was capable of modelling the time series with strong multiple seasonalities at?

- A. day level
- B. week level
- C. year level
- D. All of the above

Ans : . All of the above

51. Which of the following is the square root of the mean square error?

- A. AMSQ
- B. RMSE
- C. AMSE
- D. RMES

Ans : . RMSE

52. Mean Absolute Error is the average of absolute difference between?

- A. normal values and true values
- B. predicted values and imaginary values
- C. predicted values and true values
- D. imaginary values and true values

Ans : . predicted values and true values

53. Noise is the error in the observations added due to environmental factors.

- A. Yes

- B. No
- C. Can be yes or no
- D. Can not say

Ans : . Yes

54. What is the dominant modality for communication between humans?

- a) Hear
- b) Speech
- c) Smell
- d) None of the mentioned

Answer: Speech

55. What kind of signal is used in speech recognition?

- a) Electromagnetic signal
- b) Electric signal
- c) Acoustic signal
- d) Radar

Answer: Acoustic signal

56. What is viewed as problem of probabilistic inference?

- a) Speech recognition
- b) Speaking
- c) Hearing
- d) Utterance

Answer: Speech recognition

57. Which specifies the prior probability of each utterance?

- a) Sound model
- b) Model
- c) Language model
- d) All of the mentioned

Answer: Language model

58. Which model gives the probability of each word following each other word?

- a) Bigram model
- b) Diagram model
- c) Gram model
- d) Speech model

Answer: Bigram model

59. What is the study of how the language sounds?

- a) Speechology
- b) Biology
- c) Trilogy
- d) Phonology

Answer: Phonology

60. What are periodic changes in pressure that propagate through the air?

- a) Air waves
- b) Sound waves
- c) Rate
- d) None of the mentioned

Answer: Sound waves

61. What is called as the properties of the signal that extend over interval?

- a) Hops
- b) Rate
- c) Frames
- d) All of the mentioned

Answer: Frames

62. Which is used to capture the internal structure of the phones?

- a) One-state phone model
- b) Two-state phone model
- c) Three-state phone mone
- d) All of the mentioned

Answer: Three-state phone mone

63. Which are partially captured by triphone model?

- a) Articulation effects

- b) Coarticulation effects
- c) Both Articulation & Coarticulation effects
- d) None of the mentioned

Answer: Coarticulation effects

64. What is the other name of informed search strategy?

- a) Simple search
- b) Heuristic search
- c) Online search
- d) None of the mentioned

Answer: Heuristic search

65. How many types of informed search method are in artificial intelligence?

- a) 1
- b) 2
- c) 3
- d) 4

Answer: 4

66. Which search uses the problem specific knowledge beyond the definition of the problem?

- a) Informed search
- b) Depth-first search
- c) Breadth-first search
- d) Uninformed search

Answer: Informed search

67. Which function will select the lowest expansion node at first for evaluation?

- a) Greedy best-first search
- b) Best-first search
- c) Depth-first search
- d) None of the mentioned

Answer: Best-first search

68. What is the heuristic function of greedy best-first search?

- a) $f(n) \neq h(n)$
- b) $f(n) < h(n)$

c) $f(n) = h(n)$

d) $f(n) > h(n)$

Answer: $f(n) = h(n)$

69. Which search uses only the linear space for searching?

a) Best-first search

b) Recursive best-first search

c) Depth-first search

d) None of the mentioned

Answer: Recursive best-first search

70. Which method is used to search better by learning?

a) Best-first search

b) Depth-first search

c) Metalevel state space

d) None of the mentioned

Answer: Metalevel state space

71. Which search is complete and optimal when $h(n)$ is consistent?

a) Best-first search

b) Depth-first search

c) Both Best-first & Depth-first search

d) A* search

Answer: A* search

72. Which is used to improve the performance of heuristic search?

a) Quality of nodes

b) Quality of heuristic function

c) Simple form of nodes

d) None of the mentioned

Answer: Quality of heuristic function

73. Which search method will expand the node that is closest to the goal?

a) Best-first search

b) Greedy best-first search

c) A* search

d) None of the mentioned

Answer: Greedy best-first search

74. Which of the following is a subset of machine learning?

- A. Numpy
- B. SciPy
- C. Deep Learning
- D. All of the above

Ans : Deep Learning

75. How many layers Deep learning algorithms are constructed?

- A. 2
- B. 3
- C. 4
- D. 5

Ans : 3

76. The first layer is called the?

- A. inner layer
- B. outer layer
- C. hidden layer
- D. None of the above

Ans : inner layer

77. RNNs stands for?

- A. Receives neural networks
- B. Report neural networks
- C. Recording neural networks
- D. Recurrent neural networks

Ans : Recurrent neural networks

78. Which of the following is/are Common uses of RNNs?

- A. BusinessesHelp securities traders to generate analytic reports
- B. Detect fraudulent credit-card transaction

- C. Provide a caption for images
- D. All of the above

Ans : All of the above

79. Which of the following is well suited for perceptual tasks?

- A. Feed-forward neural networks
- B. Recurrent neural networks
- C. Convolutional neural networks
- D. Reinforcement Learning

Ans : Convolutional neural networks

80. CNN is mostly used when there is an?

- A. structured data
- B. unstructured data
- C. Both A and B
- D. None of the above

Ans : unstructured data

81. What is the primary goal of computer vision?

- A) To create software that can process natural language
- B) To enable machines to interpret and understand visual information from the world
- C) To develop algorithms for data encryption
- D) To build efficient databases for large-scale data storage

Answer: To enable machines to interpret and understand visual information from the world

82. Which of the following is a common application of computer vision?

- A) Text classification
- B) Image recognition
- C) Speech synthesis
- D) Data compression

Answer: Image recognition

83. In computer vision, what does the term “object detection” refer to?

- A) Identifying the type of an object within an image
- B) Finding and locating objects within an image
- C) Converting text into speech
- D) Analyzing the sentiment of a text

Answer: Finding and locating objects within an image

84. Which algorithm is commonly used for object detection?

- A) Support Vector Machines (SVM)
- B) Convolutional Neural Networks (CNNs)
- C) K-Means Clustering
- D) Principal Component Analysis (PCA)

Answer: Convolutional Neural Networks (CNNs)

85. What is the purpose of the “Convolutional Layer” in a CNN?

- A) To perform downsampling
- B) To extract features from the input image
- C) To merge multiple features
- D) To flatten the input data

Answer: To extract features from the input image

86. Which of the following techniques is used for image classification?

- A) Histogram of Oriented Gradients (HOG)
- B) Scale-Invariant Feature Transform (SIFT)
- C) Support Vector Machines (SVM)
- D) Optical Character Recognition (OCR)

Answer: Support Vector Machines (SVM)

87. What does “Segmentation” refer to in computer vision?

- A) Identifying the main object in an image
- B) Dividing an image into multiple segments or regions
- C) Extracting features from an image
- D) Classifying objects in an image

Answer: Dividing an image into multiple segments or regions

88. Which model architecture is known for its deep layers and is commonly used in computer vision tasks?

- A) Decision Trees
- B) Random Forest
- C) Convolutional Neural Networks (CNNs)
- D) K-Nearest Neighbors (KNN)

Answer: Convolutional Neural Networks (CNNs)

89. Produces two new offspring from two parent string by copying selected bits from each parent is called

- (A) Mutation
- (B) Inheritance
- (C) Crossover
- (D) None of these

Ans: Crossover

90. Genetic operators includes

- (A) Crossover
- (B) Mutation
- (C) Both A & B
- (D) None of these

Ans: Both A & B

91. What is the correct representation of GA?

- (A) GA(Fitness, Fitness_threshold, p)
- (B) GA(Fitness, Fitness_threshold, p, r)
- (C) GA(Fitness, Fitness_threshold, p, r, m)
- (D) GA(Fitness, Fitness_threshold)

Ans: GA(Fitness, Fitness_threshold, p, r, m)

92. When would the genetic algorithm terminate?

- (A) Maximum number of generations has been produced
- (B) Satisfactory fitness level has been reached for the population.
- (C) Both A & B
- (D) None of these

Ans: Both A & B

93. Genetic algorithm is a

- (A) Search technique used in computing to find true or approximate solution to optimization and search problem*
- (B) Sorting technique used in computing to find true or approximate solution to optimization and sort problem*
- (C) Both A & B*
- (D) None of these*

Ans: *Search technique used in computing to find true or approximate solution to optimization and search problem*

94. GA techniques are inspired by _____ biology.

- (A) Evolutionary*
- (B) Cytology*
- (C) Anatomy*
- (D) Ecology*

Ans: *Evolutionary*

95. _____ evolution over many generations was directly influenced by the experiences of individual organisms during their lifetime

- (A) Baldwin
- (B) Lamarckian
- (C) Bayes

(D) None of these

Ans: Lamarckian

96. How many types of feedback does reinforcement provide?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: 2

97. Which kind of data does reinforcement learning use?

- A. Labeled data
- B. Unlabelled data
- C. None
- D. Both

Answer: None

98. Reinforcement learning methods learned through ____?

- A. Experience
- B. Predictions
- C. Analyzing the data

Answer: Experience

99. How many types of machine learning are there?

- A. 2
- B. 3
- C. 4

D. 5

Answer: 4

100. Which of the following is the practical example of reinforcement learning?

- A. House pricing prediction
- B. Market basket analysis
- C. Text classification
- D. Driverless cars

Answer: Driverless cars

1. Which of the following is used to define a block of code in Python?

- A) Parentheses
- B) Indentation
- C) Curly braces
- D) Colons

Answer: B) Indentation

2. What is the output of the following Python code?

`python`

`Copy`

```
x = "Hello"
```

```
y = "World"
```

```
print(x + y)
```

- A) Hello World
- B) Hello
- C) World
- D) HelloWorld

Answer: D) HelloWorld

3. Which function can be used to get the length of a list in Python?

- A) len()
- B) length()
- C) size()
- D) count()

Answer: A) len()

4. What does the following Python code do?

python

Copy

```
x = [1, 2, 3]
```

```
y = x.copy()
```

```
y[0] = 100
```

```
print(x)
```

- A) [100, 2, 3]
- B) [1, 2, 3]
- C) [100, 1, 2]
- D) Error

Answer: B) [1, 2, 3]

5. Which of the following is the correct way to handle exceptions in Python?

- A) try-except-finally
- B) try-finally
- C) catch-finally
- D) try-catch

Answer: A) try-except-finally

11. Which operator is used for exponentiation in Python?

- A) *
- B) /
- C) **
- D) ^

Answer: C) **

12. What will the following code output?

python

Copy

```
x = [1, 2, 3, 4]
```

```
x.remove(3)
```

```
print(x)
```

- A) [1, 2, 4]
- B) [1, 2, 3]
- C) [3, 4]
- D) Error

Answer: A) [1, 2, 4]

13. What is the output of the following code?

python

Copy

```
x = 'Python'
```

```
print(x[2:5])
```

- A) Py
- B) tho
- C) tho
- D) t

Answer: B) tho

14. Which of the following is the correct syntax to define a function in Python?

- A) `def function():`
- B) `function() =>`
- C) `define function():`
- D) `function = def():`

Answer: A) `def function():`

15. Which method is used to remove all elements from a list in Python?

- A) `clear()`
- B) `delete()`
- C) `pop()`
- D) `remove()`

Answer: A) `clear()`